

Chapter 16. The Integral in n -Space

1. The definition of **double integral** over a closed rectangle $\mathbf{R}$ (Page 686) and the Integrability Theorem (Theorem A, Page 687).

2. Let the rectangle $\mathbf{R} = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$, then

$$\iint_{\mathbf{R}} f(x, y) \, dA = \int_c^d \left[\int_a^b f(x, y) \, dx \right] dy = \int_a^b \left[\int_c^d f(x, y) \, dy \right] dx$$

The last two expressions are called **iterated integrals** (Page 692).

3. Let S be a nonrectangular region in the plane (Page 695-696).

- $\mathbf{S}$ is an **y-simple** set such as

$$\mathbf{S} = \{(x, y) \mid a \leq x \leq b, \phi_1(x) \leq y \leq \phi_2(x)\}$$

Then

$$\iint_{\mathbf{S}} f(x, y) \, dA = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y) \, dy dx$$

- $\mathbf{S}$ is an **x-simple** set such as

$$\mathbf{S} = \{(x, y) \mid c \leq y \leq d, \psi_1(y) \leq x \leq \psi_2(y)\}$$

Then

$$\iint_{\mathbf{S}} f(x, y) \, dA = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y) \, dx dy$$

4. The area of a region $\mathbf{S}$ in the plane can be calculated by

$$A(\mathbf{S}) = \iint_{\mathbf{S}} dA.$$

5. If $f(x, y) \geq 0$, the volume of the solid under the surface $z = f(x, y)$ and above the region $\mathbf{S}$ is given by

$$V = \iint_{\mathbf{S}} f(x, y) \, dA.$$

6. A polar rectangle $\mathbf{R}$ has the form

$$\mathbf{R} = \{(r, \theta) \mid a \leq r \leq b; \alpha \leq \theta \leq \beta\}.$$

The double integral in polar coordinates over $\mathbf{R}$ (Page 702):

$$\iint_{\mathbf{R}} f(x, y) \, dA = \iint_{\mathbf{R}} f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta$$

7. Let S be a general region in the plane (Page 703).

- $\mathbf{S}$ is an **r-simple** set such as

$$\mathbf{S} = \{(x, y) \mid \phi_1(\theta) \leq r \leq \phi_2(\theta), \alpha \leq \theta \leq \beta, \}$$

Then

$$\iint_{\mathbf{S}} f(x, y) \, dA = \int_{\alpha}^{\beta} \int_{\phi_1(\theta)}^{\phi_2(\theta)} f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta$$

- $\mathbf{S}$ is an **θ -simple** set such as

$$\mathbf{S} = \{(x, y) \mid a \leq r \leq b, \psi_1(r) \leq \theta \leq \psi_2(r)\}$$

Then

$$\iint_{\mathbf{S}} f(x, y) \, dA = \int_a^b \int_{\psi_1(r)}^{\psi_2(r)} f(r \cos \theta, r \sin \theta) \, r \, d\theta \, dr$$

8. (Page 707) Given a lamina covers a region $\mathbf{S}$ in the xy -plane with a density $\delta(x, y)$, then its total mass m is given by

$$m = \iint_{\mathbf{S}} \delta(x, y) \, dA$$

The center of mass of the lamina $(\bar{x}, \bar{y})$ is given by

$$\bar{x} = \frac{M_y}{m} \quad \text{where} \quad M_y = \iint_{\mathbf{S}} x \delta(x, y) \, dA,$$

$$\bar{y} = \frac{M_x}{m} \quad \text{where} \quad M_x = \iint_{\mathbf{S}} y \delta(x, y) \, dA.$$

9. Suppose that G is the surface defined by $z = f(x, y)$ and over the closed and bounded region $\mathbf{S}$, then the area of G is (Page 712)

$$A(G) = \iint_{\mathbf{S}} \sqrt{f_x^2 + f_y^2 + 1} \, dA,$$

10. Triple integrals in Cartesian coordinates (Page 716-717). Let $\mathbf{B}$ be a box in three space such as

$$\mathbf{B} = \{(x, y, z) \mid a_1 \leq x \leq a_2, \, b_1 \leq y \leq b_2, \, c_1 \leq z \leq c_2\},$$

then

$$\iiint_{\mathbf{B}} f(x, y, z) \, dV = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z) \, dz dy dx$$

Let $\mathbf{S}$ be a closed bounded set in three-space. Assume that $\mathbf{S}$ is z -simple and its projection in xy -plane is y -simple such as

$$\mathbf{S} = \{(x, y, z) \mid a_1 \leq x \leq a_2, \, \phi_1(x) \leq y \leq \phi_2(x), \, \psi_1(x, y) \leq z \leq \psi_2(x, y)\},$$

then

$$\iiint_{\mathbf{S}} f(x, y, z) \, dV = \int_{a_1}^{a_2} \int_{\phi_1(x)}^{\phi_2(x)} \int_{\psi_1(x, y)}^{\psi_2(x, y)} f(x, y, z) \, dz dy dx$$

11. (Page 719) Given a solid covers a region $\mathbf{S}$ in the three-space with a density $\delta(x, y, z)$, then its total mass m is given by

$$m = \iiint_{\mathbf{S}} \delta(x, y, z) \, dV$$

The center of mass of the lamina $(\bar{x}, \bar{y}, \bar{z})$ is given by

$$\bar{x} = \frac{M_{yz}}{m} \quad \text{where} \quad M_{yz} = \iiint_{\mathbf{S}} x \delta(x, y, z) \, dV,$$

$$\bar{y} = \frac{M_{zx}}{m} \quad \text{where} \quad M_{zx} = \iiint_{\mathbf{S}} y \delta(x, y, z) \, dV,$$

$$\bar{z} = \frac{M_{xy}}{m} \quad \text{where} \quad M_{xy} = \iiint_{\mathbf{S}} z \delta(x, y, z) \, dV.$$

12. Triple integrals in Cylindrical coordinates (r, θ, z) (Page722). Let $\mathbf{S}$ be a closed bounded set in three-space. Assume that $\mathbf{S}$ is z -simple and its projection in xy -plane is r -simple such as

$$\mathbf{S} = \{(r, \theta, z) \mid r_1(\theta) \leq r \leq r_2(\theta), \theta_1 \leq \theta \leq \theta_2, g_1(r, \theta) \leq z \leq g_2(r, \theta)\},$$

then

$$\iiint_{\mathbf{S}} f(x, y, z) \, dV = \int_{\theta_1}^{\theta_2} \int_{r_1(\theta)}^{r_2(\theta)} \int_{g_1(r, \theta)}^{g_2(r, \theta)} f(r \cos \theta, r \sin \theta, z) \, r \, dz \, dr \, d\theta.$$

13. Triple integrals in Spherical coordinates (ρ, θ, ϕ) (Page724). Let $\mathbf{S}$ be a closed bounded set in three-space such as

$$\mathbf{S} = \{(\rho, \theta, \phi) \mid \rho_1 \leq \rho \leq \rho_2, \theta_1 \leq \theta \leq \theta_2, \phi_1 \leq \phi \leq \phi_2\},$$

then

$$\iiint_{\mathbf{S}} f(x, y, z) \, dV = \int_{\phi_1}^{\phi_2} \int_{\theta_1}^{\theta_2} \int_{\rho_1}^{\rho_2} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \, \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi.$$